PRARTHANA U

+91 6364608307 · prarthana6710@gmail.com · India · linkedin.com/in/prarthana-u-560833380
github.com/prarthana2031 · leetcode.com/u/prarthana127

SUMMARY

Software engineering experience in agent evaluation infrastructure. Designed multi-domain coding tasks and verification harnesses for frontier coding agents (DeepSeek-V4-Pro, Opus-4.8) under the Harbor/Dynamo format. Built adversarial benchmarks spanning PyTorch LoRA fine-tuning, multi-object tracking, ETL/CDC gates, and CI quality systems — with trusted oracles, anti-bypass tests, Dockerized environments, and GitHub Actions merge gates. Comfortable analyzing agent failure modes and turning near-miss trajectories into harder tasks.

EDUCATION

B.E. in Computer Science · Dr. Ambedkar Institute of Technology, Bengaluru — Expected 2028

EXPERIENCE

LLM Trainer · Agent Evaluation Infrastructure · Handshake AI — Jul 2026 – Sep 2026

• Designed multi-domain Harbor/Dynamo evaluation tasks for frontier coding agents (DeepSeek-V4-Pro, Opus-4.8) across PyTorch LoRA fine-tuning, multi-object tracking (IoU association, track lifecycle), CDC/ETL batch gates, and CI flake/quarantine systems — each structured as a real multi-file PR with interdependent policy bugs, human-written instructions, Docker environments, and task.toml metadata.

• Implemented agent-hard defects and trusted pytest oracles with anti-bypass safeguards (no skips/tautologies, determinism on double-run). Examples: train/eval leakage, prompt-token masking (-100), wrong LoRA injection targets, full state_dict vs A/B-only adapter checkpoints, held-out CE/accuracy contracts, first-fit vs global IoU matching, concurrent index RMW without locks, TTL unit mismatches, residual-failure logic.

• Shipped tasks through Dockerized environments and centralized GitHub Actions validation (build, oracle, review, trials). Analyzed near-miss agent trajectories to harden bug injection and acceptance criteria. Contributed 10+ accepted and merged PRs across software engineering, data processing, computer vision, and model-training domains.

• Diagnosed and repaired a broken multi-package Python monorepo CI/CD pipeline (Poetry, Docker, GitHub Actions); resolved dependency conflicts, circular imports, build-order failures, and release automation issues.

• Wrote end-to-end integration tests for distributed FastAPI-style service layouts covering cascading timeout/retry failures, auth bypass edge cases, middleware ordering, header propagation, and service-discovery faults.

PROJECTS

Watchtower — Browser Automation, MCP Server & QA Tool for AI Agents · TypeScript · Playwright · Cheerio · SQLite · Pixelmatch · Node.js · MCP — github.com/prarthana2031/Watchtower

• Building a dual-purpose system: (1) browser automation layer for AI agents via MCP server, and (2) automated QA / visual-regression tool. Playwright headless browser + Cheerio DOM parsing for page interaction, content extraction, and state capture. Normalized content trees hashed with SHA-256 and stored in SQLite for deterministic cross-run comparison; Pixelmatch for pixel-level UI diffs. Rate limiting and robots.txt compliance. Designed so agents can drive real browser sessions, inspect page state, and receive structured failure reports.

Amazon ML Challenge — Entity Resolution at Scale · Python · Polars · LightGBM · sentence-transformers · RapidFuzz · AWS EC2 · S3 · Parquet

• Built end-to-end entity resolution pipeline over ~20M business records (name + address) across multiple sources and countries. Designed multi-view blocking (token, structural, Indic/OCR repairs) achieving ≥98% pair recall with ≤25 candidates per entity. Engineered 30+ string, structural, and rank features; trained LightGBM rankers with calibrated probabilities optimized for macro F0.5.

• Implemented per-entity set decision (one-owner constraint + expected F0.5 selection) improving score by +0.03–0.05 over per-pair thresholds. Added cross-encoder re-ranking on uncertain pairs and zero-shot / self-training rules for an unseen country (France). Delivered reproducible codebase (Polars, LightGBM, sentence-transformers) with tests, ablations, and methodology write-up. Operated AWS EC2 + S3 for training and artifact storage.

TrustLens — Resume Screening Backend · Python · FastAPI · PostgreSQL · Supabase · Alembic · Pydantic · Postman — github.com/Trustlens07/Trustlens

• Designed and shipped a full resume-screening API: FastAPI routers with dependency injection, strict Pydantic request/response models, and resume file-upload handling. Normalized PostgreSQL schema (candidates, jobs, applications, match scores) on Supabase with version-controlled Alembic migrations. Implemented scoring/matching endpoints, error-handling middleware, and Postman suites covering happy-path + malformed-payload edge cases — focusing on clean contracts and production-ready request lifecycle.

TECHNICAL SKILLS

Languages: Python, TypeScript, C, Bash
ML / Training: PyTorch, LoRA / PEFT, LightGBM, Polars, sentence-transformers, ranking & calibration (macro F0.5), entity resolution, prompt-token masking (-100), checkpoint contracts
Evaluation & Agents: Adversarial task design, Harbor/Dynamo format, pytest oracles, anti-bypass safeguards, pass@k / attempt-limited eval, near-miss & failure-mode analysis
CV / Systems: IoU association, SORT-style multi-object tracking, track lifecycle, CDC sequence/identity/before-image validation, concurrent indexes (locks / RMW)
Infrastructure: Docker, GitHub Actions CI/CD, Linux, AWS (EC2, S3), Poetry, Git
Backend & Testing: FastAPI, pytest, Playwright, PostgreSQL, SQLite, Pydantic, Postman, integration testing

ACHIEVEMENTS

• Google Solution Challenge — built working prototype on AI response bias (problem framing, detection/mitigation approach, and demo).
• Active on LeetCode (arrays, trees, graphs, dynamic programming); participates in weekly contests.